SAMPLE

Earth's Sky: Star Catalogues

Bayer, Flamsteed, and the Naming of Stars

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Introduction

The night sky holds more stars than any eye can count, and the catalogue is how the counting is kept: a list of the stars, each with its place and its brightness and its name, the sky's whole population recorded in tables that began on clay and now live in databases of billions. The subject of this book is the great catalogues themselves, the documents that turned the sky's anonymous multitude into an addressed population, and the story they tell runs the whole length of the recorded astronomy: the first star lists of the ancient world, the coordinate catalogues of the Greek and Islamic astronomers, the telescopic surveys of the centuries of the telescope, the photographic programs that counted by the million, and the space missions that now weigh and place every star the eye can see and a billion more.

It is a story with famous names. Hipparchus, who made the first known catalogue in the West, the stars placed by their coordinates a century and a half before the common era; Ptolemy, whose Almagest carried the ancient catalogue to the medieval world; Ulugh Beg, the Samarqand astronomer-prince whose observatory's star tables were the best of their age; Tycho Brahe, whose naked-eye precision set the standard the telescope later beat; Bayer and Flamsteed, whose naming schemes still mark the charts; Argelander, whose Bonn survey counted a third of a million stars by eye and lantern; the Harvard computers, whose Henry Draper Catalogue classified the spectra of a quarter million stars; and the space age, Hipparcos and Gaia, the satellite astrometry that fixed the stars' places to the fineness of a hair's width at a hundred kilometers and measured a billion of them.

The catalogues' meaning is the astronomy's meaning in miniature: the science begins when the sky is recorded, the positions kept and compared, the changes caught, the models tested against the tables. A star catalogue is an instrument as surely as a telescope is, the one collecting the sky's light, the other collecting its facts, and the history of the catalogues is the history of the astronomy's data, the patient bookkeeping that made the discoveries possible: the precession found in the table-comparisons, the proper motions caught in the re-surveys, the parallaxes and distances measured by the space missions, and the whole modern census of the galaxy resting on the catalogues' columns.

A note on the book's neighbors, before the account begins. The star catalogue's subject touches several of the series' other volumes, and the boundaries are drawn here plainly: the constellations' own history, the figures and their myths, belongs to its own book, and this account keeps to the catalogues, the constellation-scheme appearing only as the naming's frame; the individual astronomers whose biographies fill the series' biographical volumes appear here as cataloguers, their lives' own stories told in their own books; and the exoplanet and the galactic sciences, the catalogue's modern users, are described as users, their own subjects their own books' ground. The numbers quoted, the catalogues' sizes and their limiting magnitudes, the mission dates and the positional precisions, are the record's established values, rounded for the telling and honest at the rounding. The reader who finishes these chapters holds the catalogues whole: their history, their anatomy, their science, and their standing, the sky's address book's complete account in plain words.

One more preliminary remark, the vocabulary's own plain start. The account uses its terms deliberately and defines them at first appearance: the catalogue itself, the table of the sky's population; the astrometry, the position-measuring's craft; the photometry, the brightness-measuring's; the epoch and the frame, the position's dating and the reference's definition; the designations, the naming-schemes' own bookkeeping; and the magnitude and the parallax, the two numbers the whole story turns on. The reader who meets the terms here will meet them again in their chapters, each technical word earning its place in the account's plain telling; and the reader who wants the catalogues' whole human story, the observers' and the computers' labor, will find it in the observation and the context chapters, the census's biography told alongside its technique. The account begins with the discovery era, and the vocabulary travels with the story, the words' own history part of the history the words describe.

One last preliminary note, the scale-feel the reader should carry. The numbers of the cataloguing's history run from the hundreds to the billions, and the account's honesty requires the reader's own sense of the jumps: the ancient catalogue's thousand, the telescopic surveys' hundred-thousands, the photographic millions, the space missions' billion, each step a hundredfold the last; and the account's own practice, quoting each catalogue's size at its place in the story, builds the ladder as it goes, the reader's scale-feel the numbers' own context. The magnitudes' scale-feel travels with the counts: the eye's six-magnitude sky of the few thousand visible stars, the telescope's and the plate's fainter depths, the satellite's twentieth-magnitude reach, each magnitude-step another population the record swallowed. The reader who finishes these chapters with the ladder built and the scale-feel set holds the

story's own spine, the census's growth the subject's whole arc, the sky's record from the handful to the billion.

The chapters that follow describe the catalogues whole: what they are and how they are made, the coordinates and the magnitudes, the designations and the epochs; the history from Babylon to Gaia; the science the catalogues carry, the astrometry and the photometry; the great surveys compared; the challenges of the great data, the designations' tangles and the star-sellers' fictions; and the future, the ever-larger sky-counts and the deep-sky archives. The reader should bring from the start the catalogue's own double character: the driest of documents, a table of numbers, and yet the sky's own address book, the book of the stars that the whole science keeps, and the account of its making is the narrative of how the species organized the heavens, one entry at a time, for two and a half thousand years.

Chapter 1: The Discovery of Star Catalogues

The star catalogue's discovery, in the sense of the first known making, is buried in the ancient Near East, and the story begins with the lists. The Babylonian tablets, the mul.apin series and its kin, recorded the stars by their names and their calendar-risings, the sky's order written for the purposes of the calendar and the omen-craft; the Egyptian and the Chinese records kept their own star-lists, the Chinese tradition among the most continuous, the constellations and the guest stars recorded dynasty by dynasty; and the lists' common character, the sky divided into named parts and the parts held in a record, is the catalogue's own idea in its first form: the sky that can be written down.

The first catalogue in the modern sense, the stars placed by measured coordinates, belongs to Hipparchus, the Greek astronomer of the second century before the calendar's turn, and the discovery's story is one of the astronomy's founding anecdotes. Comparing his own observations with the records of the earlier Greek astronomers, Hipparchus found the sky's longitudes shifted, the whole sphere slowly moving, the discovery of the precession; and the catalogue he compiled, the stars listed with their positions, was both the tool and the monument of the discovery, the sky's first written reference-frame, roughly eight hundred and fifty entries with brightness noted, the cataloguing's founding document. The catalogue itself is lost, its content preserved through Ptolemy's borrowing, but its existence and its character are the history's settled facts, the catalogue born as the instrument of the comparison, the data-made-discovery at the tradition's very start.

Ptolemy's Almagest, three centuries later, carried the catalogue to the medieval world: the great compendium's star tables, roughly a thousand entries in the coordinate style, the northern and the southern constellations, the magnitudes noted, the whole inherited framework systematized and transmitted; the Islamic astronomers received it and improved it, checking the positions against their own measurements, and the tradition culminating in Ulugh Beg's Samarqand observatory, whose star tables of the fifteenth century, a thousand-plus stars re-observed, were the medieval catalogue's finest, the accuracy bettered only by the telescopic age.

A further thread of the discovery era deserves its paragraph, the southern sky's own late arrival, the hemisphere the ancient catalogues never held. The Greek census stopped at the horizon the Mediterranean allowed: the far-southern stars unnamed, the constellations' figures ending at the darkness below the ship's keel, the whole southern sky outside the record for the first seventeen centuries of the cataloguing; and the telescopic age's voyages filled the gap: the southern expeditions and the island stations, Halley at St. Helena, the Lacaille survey of the Cape, the southern stars first placed in the 1750s by the French astronomer's patient zones, the new constellations filling the blanks, the catalogues' whole-sky ambition completed by the expeditions. The thread's meaning is the census's own geography: the record grew with the reach, and the sky's address book, complete in principle only in the voyaging age, carries the expeditions' own history in its southern pages, the hemisphere's blank filled as the ships and the instruments extended the bookkeeping's map.

The discovery era earns one further note, the catalogue's own prehistory, the record's roots below the named tradition. The sky-records before the catalogues proper: the calendar's observations, the heliacal risings noted, the season's markers kept, the sky's watch the agriculture's own clock; the omen-records' star-lists, the Babylonian series' named stars and their risings, the astrological archive the astronomical record's own ancestor; and the myth-maps' constellations, the sky's figures the memory's own filing-system, the star-groups named and storied long before any coordinate measured them. The note's meaning is the record's own prehistoric continuity: the catalogue grew from the species' oldest skykeeping, and the table's columns, the name and the place and the brightness, were each in use before they were combined, the census's invention the fusion of the older habits, the sky-watch and the star-lore and the number-craft meeting in the ancient record that the chapter's story begins.

One closing note for the discovery chapter, the tradition's own unbroken line, the cataloguing's continuity held

plainly. The record's keeping never stopped: the Babylonian lists to the Greek tables, the Greek to the Islamic zijes, the zijes to the Latin translations, the translations to the Renaissance atlases, the atlases to the meridian volumes, the volumes to the plate-surveys, the surveys to the space census, each era's catalogue the next era's inheritance, the record's thread never cut by any empire's fall or any instrument's change. The note's meaning is the tradition's own standing in the science's history: the star catalogue is the astronomy's oldest continuously-kept record, the bookkeeping the one thread that runs the whole length of the species' sky-knowledge, and the discovery chapter's story, from the first lists to the first coordinates, is the thread's own anchoring, the record's birth the tradition's founding act, the keeping itself the discovery's truest meaning.

The telescopic era remade the cataloguing, and its discovery-phase runs from the first star places measured with the new instrument to the first great systematic surveys. Tycho's pre-telescopic catalogue, the thousand stars of his precision program, set the standard; the telescopes then counted fainter stars by the tens of thousands in the century's compendia; and the systematic age arrived with the surveys: Bayer's atlas with its lettering, Flamsteed's British catalogue of the Greenwich program, then the great nineteenth-century surveys, the Bonn Durchmusterung's three hundred thousand, the photographic programs' millions. The discovery of the catalogue, told whole, is therefore the record of the sky's progressive bookkeeping: from the named list to the coordinate table, from the table to the survey, from the survey to the space mission's billion-star census, and the reader who follows the chapters' account holds the whole lineage, the clay tablet to the spacecraft, the sky's records growing with each instrument's power.